

STAT 432 - Homework 03

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Question 1. Explain-to-me skill

The instruction-only skill is saved in `explain-to-me/SKILL.md` and installed in the local user skill directory, with an explicit-only invocation setting. The test prompt was: “Use `$explain-to-me` to read and explain Question 2.” The explicit-path test explained the goal, symbols, fixed design, repeated-noise workflow, and requested outputs; it pointed to the Week 3 ridge lecture and ended with one understanding check. It returned no numerical solution or solution code, so the skill was retained. The complete skill file is included at the end of this report.

Question 2. Nearly collinear predictors

(a) The simulation uses R with seed 43231. Independent normal vectors are generated in the order L, U_1, U_2, X_3, X_4 . The design is generated once, centered, and divided by each column’s root mean square about its mean, using divisor $n = 100$. The correlation of X_1 and X_2 is **0.997706**; the largest and smallest singular values are **14.141480** and **0.475357**.

Writing the first two predictors’ contribution as

$$X_1\hat{\beta}_1 + X_2\hat{\beta}_2 = \frac{X_1 + X_2}{2}(\hat{\beta}_1 + \hat{\beta}_2) + \frac{X_1 - X_2}{2}(\hat{\beta}_1 - \hat{\beta}_2)$$

shows why their sum is well determined but their difference is not. Since $X_1 - X_2$ is small, a large change in the coefficient difference can make only a small change in the fit. The small singular value reflects this weakly identified direction.

(b) Each of 200 repetitions draws a fresh independent training/test noise pair, reused for all three penalties. The slope estimate solves $(X^T X + 100\lambda I)\hat{\beta} = X^T(y - \bar{y})$, and predictions include the unpenalized training intercept $\bar{y}$. Standard deviations below use divisor $200 - 1$.

Quantity	$\lambda = 0$		$\lambda = 0.02$		$\lambda = 0.2$	
	Mean	SD	Mean	SD	Mean	SD
$\hat{\beta}_1$	1.4956	1.5439	1.4870	0.1615	1.3679	0.0457
$\hat{\beta}_2$	1.5098	1.5486	1.4892	0.1659	1.3687	0.0473
$\hat{\beta}_1 + \hat{\beta}_2$	3.0054	0.0951	2.9762	0.0942	2.7366	0.0865
$\hat{\beta}_1 - \hat{\beta}_2$	-0.0142	3.0911	-0.0022	0.3136	-0.0008	0.0344

λ	Mean training MSE	Mean test MSE
0	0.964756	1.062452
0.02	0.974681	1.049262
0.2	1.076760	1.133484

(c) At $\lambda = 0.02$, the coefficient difference’s SD falls from 3.0911 to 0.3136, but the sum’s SD changes only from 0.0951 to 0.0942. With little bias in the sum, average test MSE falls slightly from 1.0625 to 1.0493. At $\lambda = 0.2$, the mean sum is 2.7366 rather than the true value 3, and test MSE rises to 1.1335: stronger shrinkage adds bias. The large coefficient stabilization has a smaller effect on predictions because changes in the weak difference direction are multiplied by the small vector $X_1 - X_2$.

Question 3. What ridge shrinks

(a) Differentiating the objective gives

$$\begin{aligned}\nabla L_\lambda(\beta) &= -\frac{1}{n}X^T(\tilde{y} - X\beta) + \lambda\beta, \\ \nabla L_\lambda(\hat{\beta}_\lambda) &= 0 \implies (X^T X + n\lambda I_p)\hat{\beta}_\lambda = X^T \tilde{y}.\end{aligned}$$

For every nonzero $v \in \mathbb{R}^p$ and $\lambda > 0$,

$$v^T(X^T X + n\lambda I_p)v = \|Xv\|^2 + n\lambda\|v\|^2 > 0.$$

The coefficient matrix is positive definite and therefore invertible. The objective is strictly convex, so the solution is unique even when X is rank deficient or $p > n$.

(b) With the compact SVD $X = U_r D_r V_r^T$, the right-hand side $X^T \tilde{y}$ lies in the span of V_r . Consequently,

$$\hat{\beta}_\lambda = V_r \operatorname{diag}\left(\frac{d_j}{d_j^2 + n\lambda}\right) U_r^T \tilde{y}.$$

Multiplication by X yields

$$X\hat{\beta}_\lambda = U_r \operatorname{diag}\left(\frac{d_j^2}{d_j^2 + n\lambda}\right) U_r^T \tilde{y} = \sum_{j=1}^r \rho_j(\lambda) u_j(u_j^T \tilde{y}).$$

OLS projects the centered response onto the column space of X , giving $\sum_j u_j(u_j^T \tilde{y})$. Thus ridge multiplies the OLS fitted component in direction u_j by $\rho_j = d_j^2/(d_j^2 + n\lambda)$; it does not uniformly scale all raw coefficients.

Here $n\lambda = 100(0.01) = 1$, so

d_j	ρ_j	Retained component
10	$100/101 = 0.990099$	99.01%
2	$4/5 = 0.800000$	80.00%
0.2	$0.04/1.04 = 0.038462$	3.85%

Including the unpenalized intercept,

$$\text{df}_{\text{eff}} = 1 + \frac{100}{101} + \frac{4}{5} + \frac{1}{26} = 2.828561.$$

(c) The $d_3 = 0.2$ direction receives the strongest shrinkage. Under isotropic response noise with variance σ^2 , the variance of its fitted response component changes from σ^2 for OLS to $\rho_3^2 \sigma^2$ for ridge. In coefficient space, the OLS noise is amplified by $1/d_3$, which explains its instability.

If the true centered mean has a substantial component in direction u_3 , the same shrinkage removes genuine signal. Its fitted component has bias $(\rho_3 - 1)u_3^T \mu$. The variance reduction must be weighed against this squared bias; a weak design direction does not necessarily contain no signal. As $\lambda \rightarrow \infty$, the slopes vanish but the intercept remains, so total effective degrees of freedom approach 1, not 0.

Question 4. Ridge and optimization

(a) The eigenvalues of $A_\lambda = A_0 + 0.09I$ are 9.09 and 0.10. Therefore

$$\kappa(A_0) = \frac{9}{0.01} = 900, \quad \kappa(A_\lambda) = \frac{9.09}{0.10} = 90.9.$$

The ratio of largest to smallest curvature is much lower. The objective's elliptical contours are less elongated, so gradient descent is less restricted by the contrast between a steep direction and an almost flat direction. The contours are still not circular.

(b) Write $b = X^T \tilde{y}/n$. The gradient is $A_\lambda \beta - b$, and the minimizer satisfies $A_\lambda \hat{\beta}_\lambda = b$. Subtracting the minimizer from a gradient step gives

$$\begin{aligned} e^{(k+1)} &= \beta^{(k)} - \eta(A_\lambda \beta^{(k)} - b) - \hat{\beta}_\lambda \\ &= (I - \eta A_\lambda) e^{(k)}. \end{aligned}$$

If v is an eigenvector of A_0 with eigenvalue a , it is an eigenvector of A_λ with eigenvalue $a + \lambda$. Its error component is therefore multiplied by $1 - \eta(a + \lambda)$ at each step.

Every direction contracts precisely when $|1 - \eta(a + \lambda)| < 1$ for both eigenvalues. Since $M = 9.09$ is the largest, the required interval is

$$0 < \eta < \frac{2}{9.09} \approx 0.220022.$$

The endpoints are excluded: $\eta = 0$ does not move, and $\eta = 2/M$ gives a nondecaying sign change in the largest-curvature direction.

For $\eta = 1/M = 1/9.09 \approx 0.110011$, the factors are

Unpenalized eigenvalue a	Ridge curvature	Error multiplier
9	9.09	$1 - 9.09/9.09 = 0$
0.01	0.10	$1 - 0.10/9.09 = 0.988999$

The steep-direction error disappears after one step in exact arithmetic, while the flat-direction error decays slowly.

For $\eta = 2.1/M \approx 0.231023$, the largest-curvature factor is

$$1 - \frac{2.1}{M} M = -1.1.$$

Its magnitude exceeds one. With the specified nonzero initial component, this error alternates sign and grows as $(-1.1)^k$, so the method does not converge. The other factor is $1 - 0.21/9.09 \approx 0.976898$; contraction in that direction does not offset divergence in the first one.

(c) Increasing λ improves conditioning, but it also changes the fitted statistical model. Strong shrinkage can increase bias and underfit the signal even if optimization becomes easy. Optimization error measures distance to the chosen ridge minimizer; prediction error measures performance on unseen responses. A suitable statistical tuning rule uses training-only validation or a prediction-risk criterion, not the penalty giving the fastest optimizer.

Question 5. Real-estate prediction

(a) The two supplied CSV files were matched by `row_id`, giving 332 training observations and 82 held-out observations. The ID was excluded; all six supplied predictors were treated as quantitative. The 122-candidate grid contains OLS and $10^{-4+0.05l}$ for $l = 0, \dots, 120$.

In each validation fold, predictor means and root-mean-square scales were estimated from the other nine folds only. The response mean and ridge fit also used only those nine folds. The same transformation was then applied to the validation observations. Each fit used $X^T X + n_{\text{fit}} \lambda I$, with its own training size, and an unpenalized intercept.

For fold errors $E_1(\lambda), \dots, E_{10}(\lambda)$, the reported criterion and standard error are

$$\bar{E}(\lambda) = \frac{1}{10} \sum_{j=1}^{10} E_j(\lambda), \quad \text{SE}(\lambda) = \frac{\sqrt{\sum_{j=1}^{10} (E_j(\lambda) - \bar{E}(\lambda))^2 / 9}}{\sqrt{10}}.$$

The minimum is at $\lambda_{\min} = 0.01995262$, with mean validation MSE 82.97765 and SE 19.22991. The threshold is therefore 102.20756. The largest eligible grid value is $\lambda_{1\text{se}} = 1.58489319$, whose mean validation MSE is 101.59553.

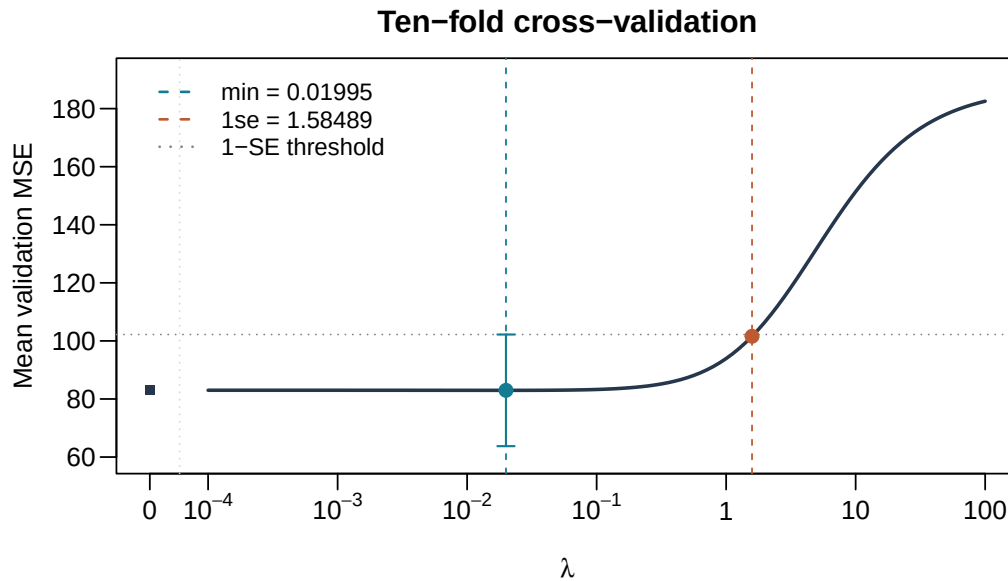


Figure 1. Ten-fold mean validation MSE over the supplied penalty grid. The OLS value is shown separately from the positive-penalty log scale. The one-SE threshold is computed using the SE at the minimum, not a different SE for each candidate.

The one-SE rule deliberately chooses the largest eligible penalty, not the penalty with the smallest mean error. It is a simplicity/stability heuristic rather than a formal confidence statement that candidate models have equal prediction risk.

(b) For GCV, predictor means and scales were estimated once from all 332 training observations. If $d_1, \dots, d_6$ are the singular values of that standardized design, the calculations used

$$\text{df}_{\text{eff}}(\lambda) = 1 + \sum_{j=1}^6 \frac{d_j^2}{d_j^2 + 332\lambda}, \quad \text{GCV}(\lambda) = \frac{\text{MSE}_{\text{train}}(\lambda)}{(1 - \text{df}_{\text{eff}}(\lambda)/332)^2}.$$

The GCV minimum occurs at $\lambda_{\text{GCV}} = 0.02818383$, with criterion value 83.50191. This criterion and its selected penalty were calculated without the test responses.

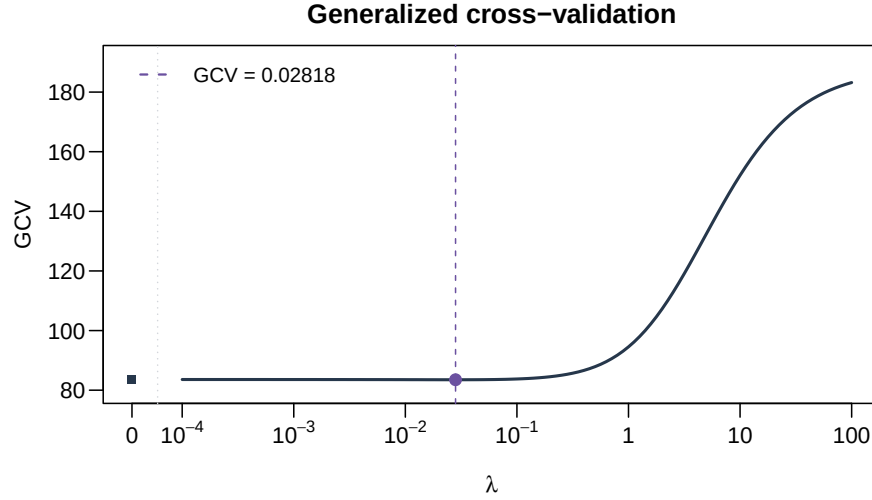


Figure 2. GCV using only the training data, with the intercept included in effective degrees of freedom. OLS is shown separately from the log scale for positive penalties.

(c) The three selected penalties were saved before evaluating the held-out responses. OLS and the three selected ridge models were then refitted on all 332 training observations. All four use the same training-derived preprocessing and are evaluated on the same 82 test observations.

Model	λ	Training MSE	Test MSE	df_{eff}
OLS	0.00000000	80.09392	66.32246	7.00000
CV minimum	0.01995262	80.13543	65.99957	6.77321
CV one-SE	1.58489319	100.37356	81.19981	3.03287
GCV	0.02818383	80.17125	65.90276	6.68866

CV-minimum and GCV select similar, modest penalties and give results close to OLS. The one-SE rule applies much stronger shrinkage, reduces effective degrees of freedom to about 3.03, and has noticeably larger training and test MSE in this split. The large SE at the CV minimum makes its eligibility threshold relatively permissive. GCV has the lowest observed test MSE in this table, but the small differences among GCV, CV-minimum, and OLS do not establish a reliable general ranking. Choosing the smallest observed test MSE after inspecting this table would turn the final test set into another tuning set and make the reported test performance optimistically selected. These results are a final comparison of the prespecified rules, not permission to retune on the test responses.

Computational details

The accompanying R scripts reproduce the numerical results and figures using the distributed data. No official solution file was used. For a clean run without an existing selection lock, use the following sequence from `homework/week-03`:

```
Rscript analysis/q2.R
Rscript analysis/q5.R select
Rscript analysis/q5.R evaluate
```

Q2 uses one fixed design and paired comparisons across penalties. Q5 first writes the training-only tuning choices, then evaluates the held-out responses in a separate phase. The selection command refuses to overwrite an existing lock. The original CSV files are retained unchanged. Reported MSE values use the number of evaluated observations as their divisor.

References

STAT 432, [Homework 03](#); [Ridge Regression: Stability Through Shrinkage](#); and [From a Penalized Objective to a Fitted Ridge Model](#).

I-Cheng Yeh, [Real Estate Valuation](#), UCI Machine Learning Repository (course-supplied observations and split assignments).

explain-to-me skill

The accompanying `agents/openai.yaml` sets `allow_implicit_invocation: false`, so the skill is not automatically applied to unrelated work. The complete contents of `explain-to-me/SKILL.md` follow:

```
---
name: explain-to-me
description: Explain a STAT 432 homework prompt when explicitly invoked; do not solve it.
---

# Intent
Help me understand what a question asks so I can attempt it myself.

- Read the requested question, not its solutions. Explain its goal in plain language.
- Define unfamiliar symbols, then list the required work and outputs by subpart.
- Separate supplied quantities, quantities to estimate, and what must stay fixed.
- Point to a relevant Week 3 lecture section: Ridge Regression: Stability Through Shrinkage, or From a Penalized Objective to a Fitted Ridge Model.
- Give no finished derivation, numerical answer, or solution code. Ask one short understanding check and wait. Do not edit or submit coursework.
```